

Franklin's Involution: Visualizing the Proof

Euler's Pentagonal Number Theorem

1. The Signed Generating Function

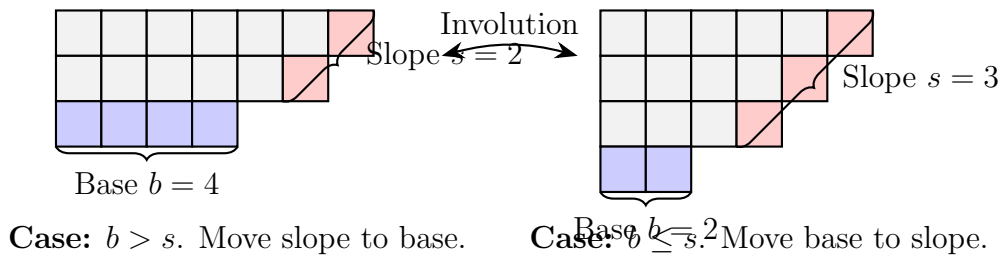
Expanding $\prod_{k=1}^{\infty} (1 - q^k)$ generates partitions into distinct parts. A partition into an even number of parts contributes $+1$, and an odd number of parts contributes -1 .

$$(1 - q)(1 - q^2)(1 - q^3) \dots = \sum (-1)^{\# \text{ parts}} q^{\text{sum of parts}}$$

2. The Sign-Reversing Involution

We pair almost all partitions by moving the **Base** b (blue) to the **Slope** s (red), or vice versa, whichever is smaller. This adds or removes exactly one part, flipping the sign and causing them to cancel in the sum.

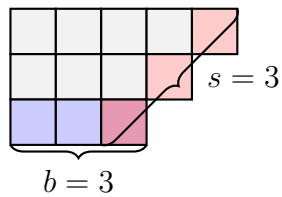
Partition: $7 + 6 + 4$ (Even, $+q^{17}$) Partition: $6 + 5 + 4 + 2$ (Odd, $-q^{17}$)



3. The Unpaired Partitions (Pentagonal Numbers)

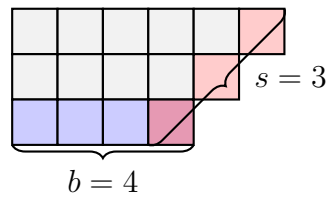
The involution fails exactly when the base and slope intersect, making it impossible to move the blocks without violating the "distinct parts" rule or creating an invalid shape. This happens in exactly two structural cases for each k .

Case 1: $b = s$ (Intersection)
 $k = 3$ parts $\implies n = \frac{k(3k-1)}{2} = 12$



Moving the base to the slope is impossible because they share a block!

Case 2: $b = s + 1$ (Intersection)
 $k = 3$ parts $\implies n = \frac{k(3k+1)}{2} = 15$



Moving the slope to the base creates a new base of length 3, under a part of length 4 (not strictly distinct!).